WASP-135b

R-0115

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-135_250216` · created 2026-07-16T04:29:24+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460722.9871 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.027
Transit depth (Rp/Rs)^2	3.03 +/- 0.93 [%]
Orbital Inclination (inc)	86.96 +/- 1.81
Airmass coefficient 1 (a1)	1.003 +/- 0.022
Airmass coefficient 2 (a2)	0.003 +/- 0.029
Scatter in the residuals of the lightcurve fit is	2.21 %
Best Comparison Star	#3 - [346, 264]
Optimal Aperture	2.42
Optimal Annulus	11.1
Transit Duration (day)	0.0682 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.027 vs 0.139 (deviation 0.99σ)
Duration fit vs published	0.0682 d vs 0.0687 d (deviation 0.06σ)
Mid-time O–C	-9.6 min
Depth SNR	4.9
Residual scatter	2.21 %

Light curve

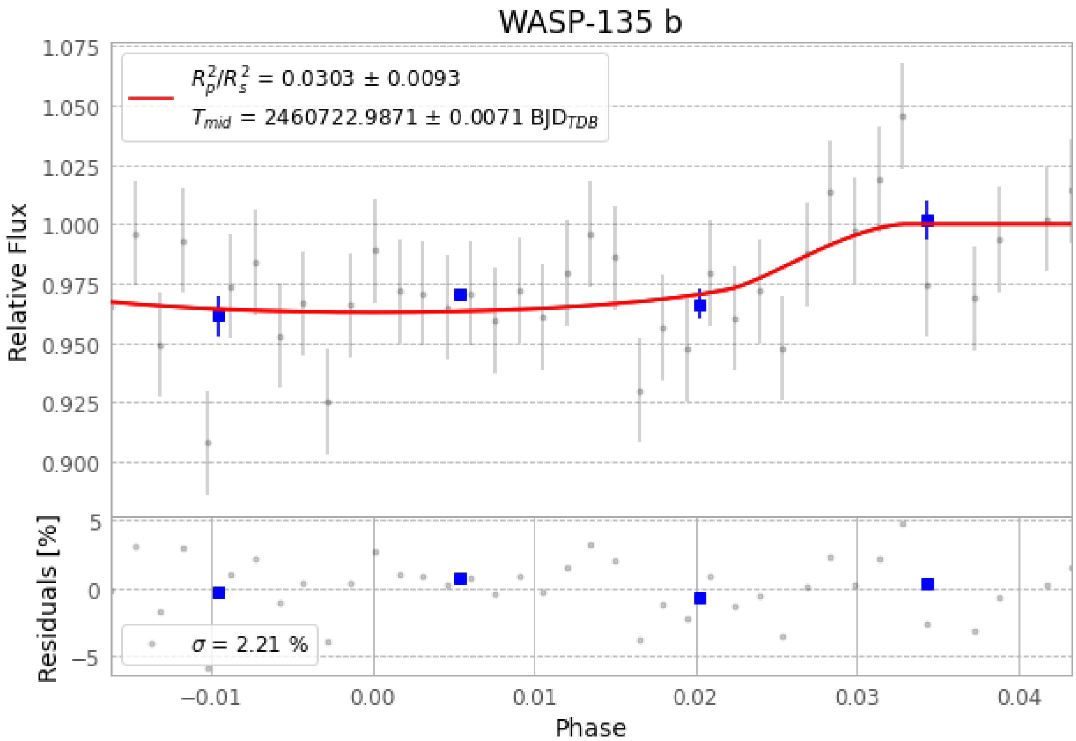

Figure 1 — FinalLightCurve_WASP-135 b_2025-02-16.png (SHA-256 b9107cc372c7fea1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 185], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 02a7aa9e23621ad2…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7`

Data provenance

41 FITS frames (27 MB), `WASP-135250216110915.FITS` → `WASP-135250216130915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250216.log (7c2545e00004…), FinalLightCurve_WASP-135 b_2025-02-16.png (b9107cc372c7…), FinalLightCurve_WASP-135 b_2025-02-16.csv (60f8e562dd20…)

Compute resources

Wall time 80.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 04:29Z.